Introduction to Mathematics and Optimisation

Exercises Week 9

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1 Exercise 7.23

We will prove later that a differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$ is strictly convex if the so-called Hessian matrix given by

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x^2}(v) & \frac{\partial^2 f}{\partial x \partial y}(v) \\ \frac{\partial^2 f}{\partial y \partial x}(v) & \frac{\partial^2 f}{\partial y^2}(v) \end{pmatrix}$$

is positive definite for every $v \in \mathbb{R}^2$. This is a multivariable generalization of the fact that $g : \mathbb{R} \to \mathbb{R}$ is strictly convex if $g''(x) > 0$ for every $x \in \mathbb{R}$.

Now let

$$f(x, y) = x^2 + y^2 - \cos(x) - \sin(y). \tag{7.12}$$

(a) Show that f is strictly convex.

(b) Compute the critical point(s) of f.

(c) For a differentiable convex function $f : \mathbb{R}^2 \to \mathbb{R}$ we have in general that

$$f(v) \geq f(u) + \nabla f(u)(v - u) \tag{7.13}$$

for every $u, v \in \mathbb{R}^2$. This is a multivariable generalization of Theorem 6.58.

Explain how one can use (7.13) to find a global minimum for the function f in (7.12). Is this minimum unique? Is $f(x, y) \geq -1$ for every $x, y \in \mathbb{R}$?

1.1 Strict Convexity

To prove strict convexity of f we need to show that Hessian matrix of f is positive definite

$$\begin{pmatrix} cos(x) + 2 & 0 \\ 0 & sin(y) + 2 \end{pmatrix}$$

By results of Exercise 3.42 this matrix is positive definite if $cos(x)+2 > 0$ and $sin(y)+2 > 0$ for all x and y. Since the minimum value of both $cos(x)$ and $sin(x)$ is -1 both of these inequities will hold true for all x and y. Therefore f is strictly convex

1.2 Roots

The critical point can be found by the following sage code

```
f(x,y) = x^2+y^2-cos(x)-sin(y)

dx(x) = f.diff(x)
dy(y) = f.diff(y)
dxx = dx.diff(x)
dxy = dx.diff(y)
dyy = dy.diff(y)

print(matrix([[dxx, dxy], [dxy, dyy]]))

def newthon(f, v):
    v_n = 0.5
    fp(v) = diff(f, v)
    while True:
        step = f(v_n) / fp(v_n)
        v_n = v_n - step
        if abs(step) == 0: # I love floats
            break
    return v_n

rx = newthon(dx, x)
ry = newthon(dy, y)

# Critical point
print([x,y], [rx, ry])

print(f, f(rx, ry))
print(f.diff())
```

$$[x, y] = [0, 0.450183611294874]$$

1.3 Global Minimum

Similarly as in Exercise 6.59 we can substitute the critical point to the inequity and since $\nabla f(u_0) = 0$ the inequity simplifies to:

$$f(v) \geq f(u_0)$$

This tells us that for any $v \in \mathbb{R}^2$ $f(v)$ is larger then our critical point therefore the minimum is unique. [1]

[1] Since $f(u_0) \approx -1.23$ $f(x, y) > -1$ is not true for all $x, y \in \mathbb{R}$